CO1-AT2-SHORT ANSWER TEST

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Course: Mobile Computing

1}

Total spectrum = 45 MHz, channel bandwidth = 25 kHz.

$$\text{Simplex channels} = \frac{45 \times 1000}{25} = 1800$$

$$\text{Full-duplex channels} = \frac{1800}{2} = 900$$

2}

Total channels = 560, cluster size = 7.

$$\text{Channels per cell} = \frac{560}{7} = 80$$

3}

100 carriers, 8 time slots per carrier.

$$100 \times 8 = 800$$

4}

Base station power = 80 W, handset power = 4 W.

$$10 \log_{10}\left(\frac{80}{4}\right) = 10 \log_{10}(20)$$

$$= 13.01 \text{ dB}$$

5}

Carrier = −60 dBm, interference = −92 dBm.

$$C/I = -60 - (-92) = 32 \text{ dB}$$

6}

Spectrum = 42 MHz, channel bandwidth = 300 kHz, cluster size = 7.

$$\text{Total channels} = \frac{42000}{300} = 140$$

$$\text{Channels per cell} = \frac{140}{7} = 20$$

7}

Initial power = 64 W, distance doubles, path-loss exponent (n=4).

$$P_r = \frac{64}{2^4} = \frac{64}{16} = 4 \text{ W}$$

8}

Inverse-square power law, (n=2).

$$\text{Power factor} = 2^2 = 4$$

9}

Cluster size = 4, channels per cell = 15

$$4 \times 15 = 60$$

11}

Frame duration = 5 ms, number of slots = 8.

$$\text{Slot duration} = \frac{5}{8} = 0.625 \text{ ms}$$

12}

Bandwidth = 24 MHz, carrier spacing = 200 kHz.

$$\frac{24000}{200} = 120$$

13}

Transmitted power = 8 W, path loss = 33 dB.

$$P_{\text{dBm}} = 10 \log_{10}(8000) = 39.03 \text{ dBm}$$

$$P_r = 39.03 - 33 = 6.03 \text{ dBm}$$

15}

6 cells, each handling 250 calls.

$$6 \times 250 = 1500$$

16}

4,000 new calls per second, average duration = 3 minutes.

$$3 \times 60 = 180 \text{ seconds}$$

$$\text{Active calls} = 4000 \times 180 = 720000$$

17}

150 channels, each requiring 25 kHz.

$$150 \times 25 = 3750 \text{ kHz}$$

$$3750/1000 = 3.75 \text{ MHz}$$

18}

30 frequency bands and 40 kHz per guard band.

$$29 \times 40 = 1160 \text{ kHz}$$

20}

8 regional centers, 40 base stations per center, 250 users per base station.

$$8 \times 40 \times 250 = 80000$$

22}

Total spectrum = 6 MHz, channel bandwidth = 30 kHz.

$$\frac{6000}{30} = 200$$

23}

1,200 subscribers, 4 calls per hour, 2.5 minutes per call.

$$A = \frac{N \times C \times T}{60}$$

$$A = \frac{1200 \times 4 \times 2.5}{60} = 200$$

24}

Bandwidth = 40 MHz, bandwidth per user = 400 kHz.

$$\frac{40000}{400} = 100$$

25}

1,200 base stations, each serving 600 users.

$$1200 \times 600 = 720000$$

26}

Total cells = 200, cells per round = 4.

$$\frac{200}{4} = 50$$

28}

600,000 subscribers, 15% roaming.

$$\frac{15}{100} \times 600000 = 90000$$

29}

Current load = 9,000 calls/s, maximum capacity = 12,000 calls/s.

$$\text{Utilization} = \frac{9000}{12000} \times 100$$

$$= 75\%$$

31}

Total bandwidth = 30 MHz, channel spacing = 200 kHz.

$$\text{Channels} = \frac{30000}{200} = 150$$